Project- High Level Design
On
MedScribe AI

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

1.1 Scope of the document

This document describes the low-level design of the MedScribe AI. It covers internal architecture, module-level design, data flow, API structure, and implementation details required to build and maintain the system.

1.2 Intended Audience

- Developers
- Testers
- Project evaluators
- Technical reviewers

1.3 System Overview

The Healthcare Content Generator is a GenAI-based application designed to generate structured medical documents such as patient summaries. The system uses Retrieval-Augmented Generation (RAG) by combining a vector database (ChromaDB) with a Large Language Model (Gemini API). It ensures consistent formatting, professional tone, and domain-specific terminology.

2. System Design

2.1 Application Design

The system follows a layered architecture consisting of three main layers: Frontend Layer, Backend Layer, and AI & Data Layer. It implements a Retrieval-Augmented Generation (RAG) pipeline to generate accurate and structured healthcare content.

Frontend Layer (User Interface)

- Built using HTML, CSS, and JavaScript
- Allows users (healthcare professionals) to input patient details
- Sends data to backend via HTTP POST (/generate)
- Displays the generated patient summary

Backend Layer (Flask)

- Acts as the core processing unit
- Handles API requests through the API Controller (app.py)
- Manages the RAG pipeline workflow
- Coordinates between embedding model, vector database, and LLM
- Uses a Prompt Engineering Module to:
    - Structure output in SOAP format
    - Maintain medical terminology and consistency

AI & Data Layer

This layer provides intelligence and context-aware generation:

- **Embedding Model (SentenceTransformer):**
    - Converts input text into vector embeddings
- **Vector Database (ChromaDB):**
    - Stores embeddings and generated documents
    - Retrieves similar past records for context (RAG)
- **LLM API (Gemini 2.5 Flash):**
    - Generates structured patient summaries
    - Uses both user input and retrieved context

Architecture:

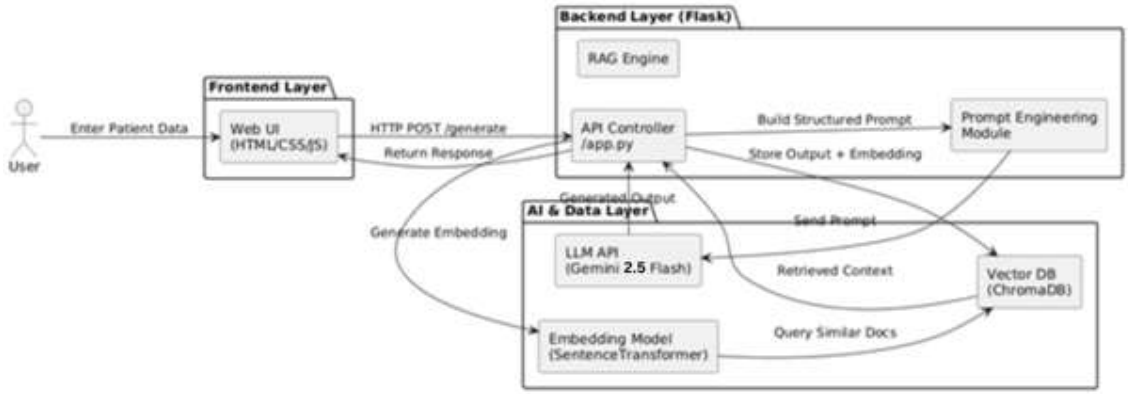
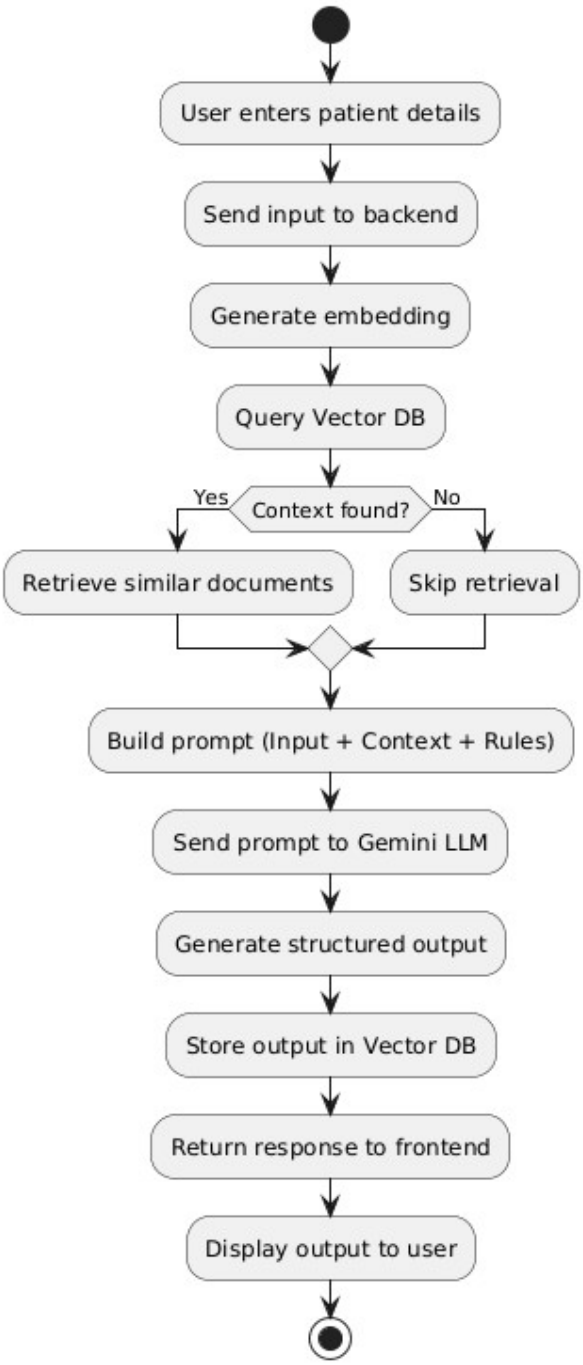


Fig. Architecture Diagram

2.2 Process Flow

Activity Diagram:



Fig, Activity Diagram

Workflow Description:

User Input:
The process starts when the user enters patient details in the frontend.

Send to Backend:
The input is sent to the Flask backend via an API request.

Embedding Generation:
The backend converts the input text into vector embeddings using the embedding model.

Vector DB Query:
The system queries the vector database to find similar past documents.

Decision – Context Found?:

- **Yes:** Relevant documents are retrieved as context
- **No:** Retrieval step is skipped

Prompt Construction:
A structured prompt is created using:

- User input
- Retrieved context (if available)
- Predefined rules (SOAP format, medical terminology)

LLM Processing:
The prompt is sent to the Gemini LLM for content generation.

Output Generation:
The LLM generates a structured patient summary.

Store Output:
The generated output is stored in the vector database for future retrieval.

Return Response:
The backend sends the response back to the frontend.

Display Output:
The final output is displayed to the user.

2.3 Information Flow

- Input Data: Patient case description
- Processed Data: Embeddings (vector format)
- Context Data: Retrieved similar documents
- Output Data: Structured patient summary (SOAP format)

Flow:

User Input → Embedding → Vector DB → Context → Prompt → LLM → Output → Storage

2.4 Components Design

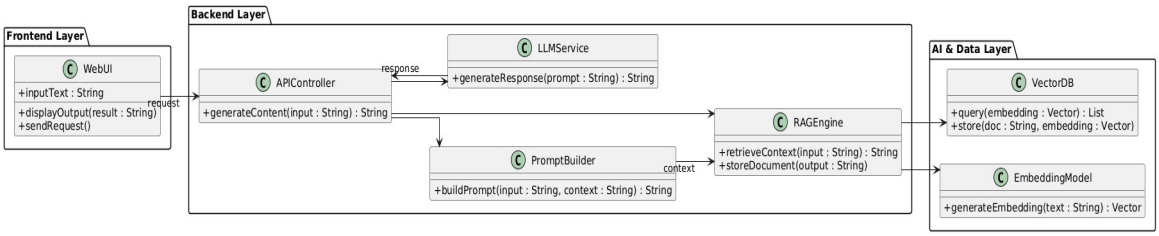


Fig. Class Diagram

Relationships Between Classes

- **WebUI → APIController:** sends user request
- **APIController → RAGEngine:** handles retrieval & storage
- **APIController → PromptBuilder:** constructs prompt
- **APIController → LLMService:** generates output
- **RAGEngine → VectorDB & EmbeddingModel:** for retrieval and storage

2.5 Key Design Considerations

Accuracy: Use RAG to improve contextual relevance

Consistency: Enforce structured format (SOAP)

Scalability: Modular backend design

Security: Avoid sensitive data exposure

Maintainability: Clean separation of modules

2.6 API Catalogue

Generate Content API

- Endpoint: /generate
- Method: POST

Request Body: Input via form

Response: Generated Summary

3. Data Design

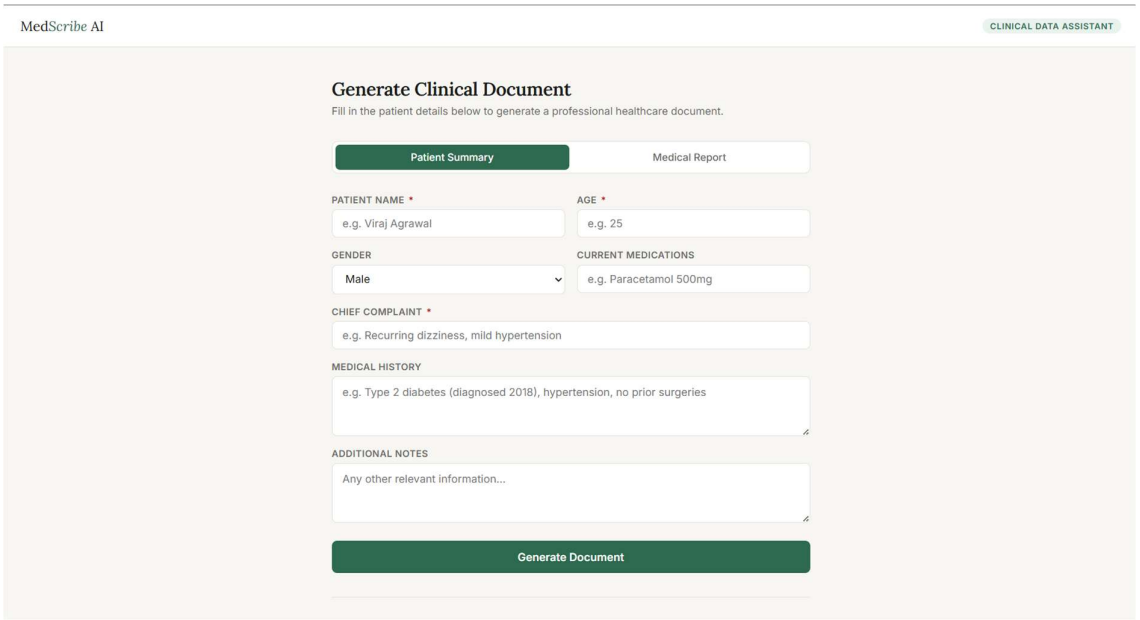
3.1 Data Model:

The system follows a three-layer architecture: a frontend presentation layer, a Python/Flask backend processing layer, and a data/Al layer. The backend orchestrates Prompt Engineering, RAG retrieval from ChromaDB, and calls to the Gemini LLM API.

3.2 Data Access Mechanism

User fills data in patient form and select document type(Summary or medical report) and then browser sends an HTTP POST request to backend, the backend then take the data and provide it to the prompt and then it goes to the Gemini 2.5 Flash model with vector embeddings and chromaDB usage and after checking for previous matching queries and then the Generated output will again be provided to the frontend interface.

4. Interface



5. State and Session Management

It is designed with a lightweight, stateless architecture that prioritizes speed and data privacy. The app treats every request as a fresh start, making it highly efficient. However, it manages state through **Retrieval-Augmented Generation (RAG)**. On the front end, simple JavaScript keeps track of what the user is currently typing and which document type they've selected. On the back end, the system uses **ChromaDB** as a sort of "long-term memory". When a user submits a case, the app quickly looks up similar past entries in this vector database to give the AI the right clinical context. While it currently runs on a fast, in-memory cache that resets with the server, it's built to easily switch to permanent storage whenever needed.

6. Non Functional Requirements

Performance: The system utilizes **ChromaDB** as an in-memory vector database to ensure near-instant retrieval of similar clinical contexts (RAG) during the generation process.

Scalability: The backend is built using **Flask**, a lightweight WSGI web application framework that allows for easy deployment.

Reliability: By integrating the **Gemini 2.5 Flash** model, the system leverages a high-availability cloud LLM API to ensure consistent document generation performance.

Usability: The frontend is designed with a clean, professional healthcare-oriented UI using **Vanilla JS, CSS, and HTML**, featuring a "one-click" generation pipeline and clear document type selectors.

Maintainability: the project follows a modular structure, separating the logic into a main backend (app.py), a clear dependency list (requirements.txt), and organized static assets (index.html).

7. References

Google Gemini API Documentation — https://ai.google.dev

ChromaDB Documentation — https://docs.trychroma.com

Flask Documentation — https://flask.palletsprojects.com

Prompt Engineering Guide — https://wmv.promptingguide.ai